01 / MOBILE / WORKSPACE STATE

Better Baseline

LIVE · Calculation service

Start with an assumption, not a prediction.

Plan a hypothetical trial. Lock enrollment. Stress-test assumptions.

SYNTHETIC PRESET / 3 EXAMPLES

Selected · Baseline collapse

20% → 10% risk · 0% → 0% dropout

`ID/version: {baseline_collapse.identity}`
`Preset hash: {baseline_collapse.hash}`

Attrition shock

20% → 20% risk · 5% → 25% dropout

`ID/version: {attrition_shock.identity}`
`Preset hash: {attrition_shock.hash}`

Stable design

20% → 19% risk · 5% → 6% dropout

`ID/version: {stable_design.identity}`
`Preset hash: {stable_design.hash}`

Lower control event risk, unchanged hypothetical effect. Synthetic assumptions; not clinical evidence.

`All presets: RR 0.75 · target 80% · two-sided α 5% · seed 42 · 10,000 replicates`

Reset to Preset Defaults

Change or reset restores preset assumptions, seed and replicates; invalidates pending work and captured results.

---

01 / Planning assumptions

Equal allocation · binary adverse outcome · two-sided superiority

Control endpoint risk (%)

20

Enter 20 for 20%. Risk in eligible control participants over the specified follow-up, not prevalence. Valid: >0 and <100.

Hypothetical effect / RR or ARR

● RR · ratio ○ ARR · pp

Switch RR / ARR with arrow keys. Hypothetical effect, not efficacy evidence.

Relative risk (ratio)

0.75

Decimal ratio; strictly >0 and <1.

Target power (%)

80

>0 and <100; must exceed alpha.

Two-sided alpha (%)

5

>0 and <100.

Dropout in each arm (%)

0

One shared value for both arms. Valid: 0 to <100.

Calculate & Lock Design

Successful calculation sets a new enrollment lock. Editing never silently replaces an existing lock.

---

02 / Enrollment lock

No live design yet

Preset inputs are not freshly calculated measurements. Calculate a design to unlock stress testing and simulation.

Awaiting calculation

Enrolled control: —
Enrolled treatment: —
Required evaluable counts: —
Analytic power: unavailable

`Source: synthetic preset`
`Follow-up: {preset.follow_up_label}`
`No clinical source asserted.`

Stress testing unavailable

LIVE ≠ clinical evidence

Fresh numerical results require the calculation service. A local backend can work without a registry or internet connection. Mode is externally configured; no mode toggle.

---

01 / INTERACTION · PRESET & RESET

Preset change/reset restores assumptions and seed/replicates, invalidates captured results and pending work. Identity/version/hash braces are binding placeholders, not fabricated identifiers. All preset options remain keyboard accessible. Power is conditional, not the probability a drug works.

02 / MOBILE / WORKSPACE STATE

Better Baseline

LIVE · Calculation service

An alternate hypothesis. The same workflow.

Plan a hypothetical trial. Lock enrollment. Stress-test assumptions.

SYNTHETIC PRESET / 3 EXAMPLES

Selected · Baseline collapse

20% → 10% risk · 0% → 0% dropout

`ID/version: {baseline_collapse.identity}`
`Preset hash: {baseline_collapse.hash}`

Attrition shock

20% → 20% risk · 5% → 25% dropout

`ID/version: {attrition_shock.identity}`
`Preset hash: {attrition_shock.hash}`

Stable design

20% → 19% risk · 5% → 6% dropout

`ID/version: {stable_design.identity}`
`Preset hash: {stable_design.hash}`

Lower control event risk, unchanged hypothetical effect. Synthetic assumptions; not clinical evidence.

`All presets: RR 0.75 · target 80% · two-sided α 5% · seed 42 · 10,000 replicates`

Reset to Preset Defaults

Change or reset restores preset assumptions, seed and replicates; invalidates pending work and captured results.

---

01 / Planning assumptions

Equal allocation · binary adverse outcome · two-sided superiority

Control endpoint risk (%)

20

Enter 20 for 20%. Risk in eligible control participants over the specified follow-up, not prevalence. Valid: >0 and <100.

Hypothetical effect / RR or ARR

RR · ratio | ARR · pp

Switch RR / ARR with arrow keys. Hypothetical effect, not efficacy evidence.

Absolute risk reduction (percentage points)

5

Enter 5 for five percentage points. Must be >0 and below control endpoint risk.

Target power (%)

80

>0 and <100; must exceed alpha.

Two-sided alpha (%)

5

>0 and <100.

Dropout in each arm (%)

0

One shared value for both arms. Valid: 0 to <100.

Calculate & Lock Design

Successful calculation sets a new enrollment lock. Editing never silently replaces an existing lock.

---

02 / Enrollment lock

Stale · prior design retained

Planning assumptions changed. Prior enrolled counts stay locked until recalculation succeeds. Stress, charts and simulation are stale.

`Prior lock retained · stale`

Control: {prior.C enrolled}
Treatment: {prior.T enrolled}
Current analytic power: unavailable

```
ARR = 5 percentage points
Source: manual user edit
Inputs: unhashed until calculated
```

Stress testing unavailable

`LIVE ≠ clinical evidence`

Fresh numerical results require the calculation service. A local backend can work without a registry or internet connection. Mode is externally configured; no mode toggle.

---

02 / INTERACTION · EDITED ARR

Only ARR is visible; RR is hidden. Manual edits acquire manual-user provenance. Retain the prior lock until recalculation succeeds; block downstream results and export. Keyboard-operable RR/ARR switch; braces are service bindings.

03 / MOBILE / WORKSPACE STATE

Better Baseline

LIVE · Calculation service

Correct the assumption before calculating.

Plan a hypothetical trial. Lock enrollment. Stress-test assumptions.

SYNTHETIC PRESET / 3 EXAMPLES

Selected · Baseline collapse

20% → 10% risk · 0% → 0% dropout

ID/version: {baseline_collapse.identity}
Preset hash: {baseline_collapse.hash}

Attrition shock

20% → 20% risk · 5% → 25% dropout

ID/version: {attrition_shock.identity}
Preset hash: {attrition_shock.hash}

Stable design

20% → 19% risk · 5% → 6% dropout

ID/version: {stable_design.identity}
Preset hash: {stable_design.hash}

Lower control event risk, unchanged hypothetical effect. Synthetic assumptions; not clinical evidence.

All presets: RR 0.75 · target 80% · two-sided α 5% · seed 42 · 10,000 replicates

Reset to Preset Defaults

Change or reset restores preset assumptions, seed and replicates; invalidates pending work and captured results.

---

01 / Planning assumptions

Equal allocation · binary adverse outcome · two-sided superiority

Control endpoint risk (%)

20

Enter 20 for 20%. Risk in eligible control participants over the specified follow-up, not prevalence. Valid: >0 and <100.

Hypothetical effect / RR or ARR

RR · ratio | ARR · pp

Switch RR / ARR with arrow keys. Hypothetical effect, not efficacy evidence.

Absolute risk reduction (percentage points)

20

ARR must be >0 and below 20 percentage points. Focused first invalid field; value retained.

Target power (%)

80

>0 and <100; must exceed alpha.

Two-sided alpha (%)

5

>0 and <100.

Dropout in each arm (%)

0

One shared value for both arms. Valid: 0 to <100.

Calculate unavailable · fix ARR

Successful calculation sets a new enrollment lock. Editing never silently replaces an existing lock.

---

02 / Enrollment lock

Invalid planning inputs

Unable to calculate. ARR cannot equal or exceed baseline risk. Values are preserved; correct the ARR field and retry.

Previous lock · stale

Control enrolled: {previous.control_enrolled}
Treatment enrolled: {previous.treatment_enrolled}
Power: stale · not current

ARR = 20 percentage points
Source: manual user edit
Current inputs: invalid / unhashed

Stress testing unavailable

LIVE ≠ clinical evidence

Fresh numerical results require the calculation service. A local backend can work without a registry or internet connection. Mode is externally configured; no mode toggle.

---

03 / INTERACTION · VALIDATION & FOCUS

On submit, announce the correction summary and focus ARR, the first invalid field. Set aria-invalid; connect the field error with aria-describedby. No request or lock replacement. Validate all ranges and target > alpha.

Better Baseline

LIVE · Calculation service

Calculating your planned design...

Plan a hypothetical trial. Lock enrollment. Stress-test assumptions.

SYNTHETIC PRESET / 3 EXAMPLES

Selected · Baseline collapse

20% → 10% risk · 0% → 0% dropout

ID/version: {baseline_collapse.identity}
Preset hash: {baseline_collapse.hash}

Attrition shock

20% → 20% risk · 5% → 25% dropout

ID/version: {attrition_shock.identity}
Preset hash: {attrition_shock.hash}

Stable design

20% → 19% risk · 5% → 6% dropout

ID/version: {stable_design.identity}
Preset hash: {stable_design.hash}

Lower control event risk, unchanged hypothetical effect. Synthetic assumptions; not clinical evidence.

All presets: RR 0.75 · target 80% · two-sided α 5% · seed 42 · 10,000 replicates

Reset to Preset Defaults

Change or reset restores preset assumptions, seed and replicates; invalidates pending work and captured results.

01 / Planning assumptions

Equal allocation · binary adverse outcome · two-sided superiority

Control endpoint risk (%)

20

Enter 20 for 20%. Risk in eligible control participants over the specified follow-up, not prevalence. Valid: >0 and <100.

Hypothetical effect / RR or ARR

RR · ratio | ARR · pp

Switch RR / ARR with arrow keys. Hypothetical effect, not efficacy evidence.

Relative risk (ratio)

0.75

Decimal ratio; strictly >0 and <1.

Target power (%)

80

>0 and <100; must exceed alpha.

Two-sided alpha (%)

5

>0 and <100.

Dropout in each arm (%)

0

One shared value for both arms. Valid: 0 to <100.

Calculating... · submission disabled

Successful calculation sets a new enrollment lock. Editing never silently replaces an existing lock.

02 / Enrollment lock

Calculation in progress

Computing required enrollment. Entered values remain visible. No new lock exists until the response succeeds.

Loading · no values yet

Required evaluable counts: pending
Enrollment lock: pending
Analytic power: pending

Source: synthetic preset
Follow-up: {preset.follow_up_label}
No clinical source asserted.

Stress testing unavailable

LIVE ≠ clinical evidence

Fresh numerical results require the calculation service. A local backend can work without a registry or internet connection. Mode is externally configured; no mode toggle.

04 / INTERACTION · REQUEST IN FLIGHT

Disable duplicate submission; expose aria-busy and announce progress politely. If inputs change, invalidate pending work: late responses must not become current. Preserve inputs on failure. No invented percentage-complete indicator.

`05 / MOBILE / WORKSPACE STATE`

Better Baseline

`LIVE · Calculation service`

Enrollment is locked. Assumptions can move.

Plan a hypothetical trial. Lock enrollment. Stress-test assumptions.

`SYNTHETIC PRESET / 3 EXAMPLES`

Selected · Baseline collapse

20% → 10% risk · 0% → 0% dropout

```
ID/version: {baseline_collapse.identity}
Preset hash: {baseline_collapse.hash}
```

Attrition shock

20% → 20% risk · 5% → 25% dropout

```
ID/version: {attrition_shock.identity}
Preset hash: {attrition_shock.hash}
```

Stable design

20% → 19% risk · 5% → 6% dropout

```
ID/version: {stable_design.identity}
Preset hash: {stable_design.hash}
```

Lower control event risk, unchanged hypothetical effect. Synthetic assumptions; not clinical evidence.

`All presets: RR 0.75 · target 80% · two-sided α 5% · seed 42 · 10,000 replicates`

Reset to Preset Defaults

Change or reset restores preset assumptions, seed and replicates; invalidates pending work and captured results.

---

01 / Planning assumptions

Equal allocation · binary adverse outcome · two-sided superiority

Control endpoint risk (%)

20

Enter 20 for 20%. Risk in eligible control participants over the specified follow-up, not prevalence. Valid: >0 and <100.

Hypothetical effect / RR or ARR

`RR · ratio` | `ARR · pp`

Switch RR / ARR with arrow keys. Hypothetical effect, not efficacy evidence.

Relative risk (ratio)

0.75

Decimal ratio; strictly >0 and <1.

Target power (%)

80

>0 and <100; must exceed alpha.

Two-sided alpha (%)

5

>0 and <100.

Dropout in each arm (%)

0

One shared value for both arms. Valid: 0 to <100.

Recalculate & Lock Design

Successful calculation sets a new enrollment lock. Editing never silently replaces an existing lock.

---

02 / Design current

`Enrollment locked`

```
Control: {C enrolled}
Treatment: {T enrolled}
Before losses · immutable
during stress testing
```

Metric	Control	Treatment
Required evaluable	`{C required}`	`{T required}`
Planned enrolled	`{C enrolled}`	`{T enrolled}`
Expected retained[1]	`{C expected}`	`{T expected}`
Endpoint risk	`20%`	`{implied risk}`
Nominal analytic power	`{power at required n}`	`{same design power}`

[1] Floored expected retained counts; not guaranteed. Sampled actual retained counts are only available after simulation.

```
Cohen’s-h / normal approximation
Model: {model_version}
Method: {method_version}
Inputs hash: {inputs_hash}
Scenario: {scenario_id}
Follow-up: {follow_up_label}
Generator: {generator}
Source/retrieval/verification: {returned
metadata}
```

`Calculation warnings`

{service warnings, or explicit “None returned”}
Source: synthetic assumptions. No clinical source asserted.

Evaluate Stress Scenario

`05 / INTERACTION · CURRENT SERVICE RESPONSE`

LIVE success schema mockup: every brace binds to an actual service-return value, never a fabricated result. Current status requires matching inputs_hash. Successful recalculation replaces the lock and invalidates downstream captured results. Expand methods/warnings via keyboard-operable disclosure.

Better Baseline

LIVE / SERVICE BINDINGS

06 / CURRENT / Route / · Service-value bindings. Charts are illustrative scaffolds, not live calculations.

Stress-test the assumptions.

LOCKED ENROLLMENT

‹N_C› control
‹N_T› treatment

Enrolled before losses · frozen
Expected retained: floor(N × (1–d)) / arm
Stress edits never change enrollment.

Selected stress scenario

Control endpoint risk

10% / fraction 0.10
1–99% · steps of 1 percentage point

Symmetric dropout

0% / fraction 0.00
0–50% · steps of 1 percentage point

Inherited RR 0.75 · target 80%
ARR plan: keep inherited ARR.
Change effect only in planning; recalculate.

Evaluate Stress Scenario

Load grid response from backend

CURRENT · Inputs hash matches evaluation and grid. Next edit marks results stale immediately; refresh after 350ms.

Original vs selected

Control risk	Plan: 20% Stress: 10%
Dropout	Plan: 0% planned Stress: 0% selected
Frozen enrollment C / T	Plan: ‹N_C› / ‹N_T› Stress: ‹N_C› / ‹N_T›
Conditional power	Plan: ‹original power› Stress: ‹selected power›

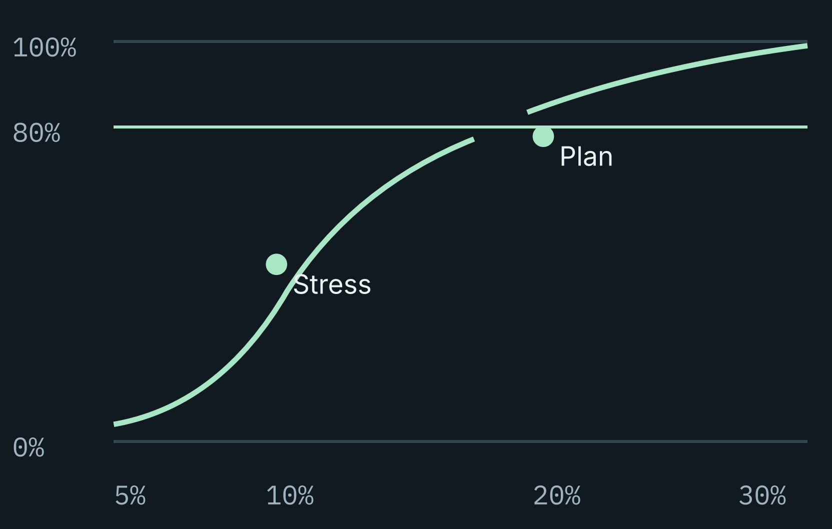
```
Expected retained C / T (floored)
Plan: ‹E_C plan› / ‹E_T plan›
Stress: ‹E_C stress› / ‹E_T stress›
```

‹Δ power› percentage points

Target 80% · Cohen's-h / normal approximation
Hash ‹matching hash› · version ‹method version›
Warnings ‹diagnostics› · null is not zero.

Conditional power curve

Fixed ‹N_C› / ‹N_T› · selected dropout exactly 0%
Schematic trace, not computed series.



Control baseline risk (%) →
Conditional analytic power (%) ↑
Reference: target 80% · gaps at invalid values

Numerical alternative / select baseline

10% · ‹power› · selected

20% · ‹power› · select baseline

‹risk› · null × · selection blocked

Original point = original risk AND planned dropout. Selected point = current risk at selected dropout. If dropout differs, original is a separate reference, not on this curve. Enter/Space selects baseline only; dropout stays fixed.

Sensitivity grid

Analytic power · risk columns / dropout rows
Target 80% boundary from backend segments

Loss↓	5%	10%	20%	30%
0%	‹p›	47%✓	‹p›	‹p›
10%	‹p›	‹p›	‹p›	‹p›
20%	‹p›	‹p›	‹p›	‹p›
30%	null×	‹p›	‹p›	‹p›

✓ selected · × invalid / unavailable
‹p› = returned analytic power, no invented series. Target boundary must not cross invalid cells.

This numerical grid has matching heatmap values. Valid cells are 44px keyboard/touch targets, announced with risk, loss and power. Select changes both controls; null cells disabled. Full precision is available in CSV.

Download analytic grid CSV

Continue below: selected-scenario simulation

Route / · Vertical order; no hidden controls or charts. Enter/Space on numerical rows selects; nulls disabled. Backend progress/error announced; Retry evaluation / Retry grid preserve edits. Charts/grid share matching data; CSV includes fractions, null flags, hashes and warnings.

Better Baseline

LIVE / SERVICE BINDINGS

07 / STALE → LOADING · Route /
Illustrative scaffold; no fresh service result.

Stress changed. Refreshing results.

`LOADING / ‹backend phase› ‹completed› / ‹total› · matching hash required`

On error: ‹diagnostic›
Retry evaluation / Retry grid. Edits and enrollment remain unchanged.

LOCKED ENROLLMENT

‹N_C› control
‹N_T› treatment

Frozen before losses. Retained = floor(N × (1–d)) / arm. Enrollment unchanged.

Selected stress scenario

Control endpoint risk

`12% / fraction 0.12`
1–99% · steps of 1 percentage point

Symmetric dropout

`0% / fraction 0.00`
0–50% · steps of 1 percentage point

Inherited RR 0.75 · target 80%
Change effect in planning, then recalculate.

Evaluate Stress Scenario · pending

Load grid response · pending

STALE → LOADING · Risk 10% → 12%.
Enrollment unchanged. Refresh after 350ms; explicit actions below dispatch the same requests. Duplicate requests disabled.

Original vs selected

Control risk	Plan: 20% Stress: 12% · was 10%
Dropout	Plan: 0% planned Stress: 0% selected
Frozen enrollment C / T	Plan: ‹N_C› / ‹N_T› Stress: ‹N_C› / ‹N_T›
Conditional power	Plan: ‹original power› Stress: ‹previous power› · STALE (10%)

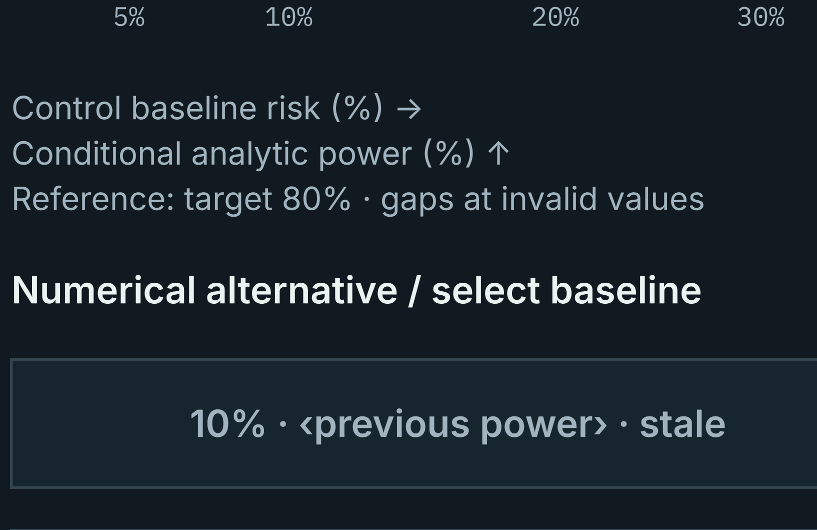
```
Expected retained C / T (floored)
Plan: ‹E_C plan› / ‹E_T plan›
Stress: ‹E_C stress› / ‹E_T stress›
Unchanged at dropout 0%.
```

Δ power unavailable while loading

Cohen's-h / normal approximation
Result ID ‹matching hash› · invalid power is null, not zero.

Power curve · stale

STALE trace · previous risk 10%; dropout 0%.
New 12% marker waits for matching response.



Control baseline risk (%) →
Conditional analytic power (%) ↑
Reference: target 80% · gaps at invalid values

Numerical alternative / select baseline

10% · ‹previous power› · stale

20% · ‹previous power› · stale

‹risk› · null × · selection blocked

Original plan point: 20% / planned dropout 0%. Curve uses selected dropout; selecting baseline preserves it. Keyboard buttons provide the same interaction.

Sensitivity grid · stale

LOADING GRID · previous values are stale.
No selection or CSV until matching response.

Loss↓	5%	10%	20%	30%
0%	‹p›	old	‹p›	‹p›
10%	‹p›	‹p›	‹p›	‹p›
20%	‹p›	‹p›	‹p›	‹p›
30%	null×	‹p›	‹p›	‹p›

Previous grid · stale / read-only
‹p› = previous returned power; new selection waits for matching hash. Target boundary never spans null.

Keyboard/touch selection and CSV disabled while stale/loading. Matching grid enables 44px valid cell targets. Invalid cells stay disabled. Errors preserve frozen enrollment and edits; Retry grid is available on failure.

Download analytic grid CSV

Simulation blocked · awaiting current result

```
Backend progress: ‹phase / completed / total›.
Error: ‹diagnostic› → Retry evaluation / Retry
grid; keep edits and frozen lock. Ignore late
responses with a different inputs hash.
Announce via polite live region.
```

Better Baseline

08 · Valid selection / simulation ready · Detail state of / — not a route

Simulate one selected scenario

LIVE / SERVICE-BINDING PREVIEW · VALID SELECTION

Selected grid point: control risk 10% · dropout 0%

FROZEN CONTEXT
Risk 10% (0.10) · dropout 0% (0.00)
Enrolled C / T: ‹N_C› / ‹N_T›
Retained: floor(N_C × (1−d)) / floor(N_T × (1−d))
Inherited RR 0.75 (or inherited ARR for ARR plan)
Target 80% · analytic power ‹matching power›
Cohen’s-h / normal approximation
Hash ‹matching inputs hash› · version ‹method version›
Warnings ‹diagnostics›

A valid grid cell updates both stress controls. A curve-table action updates baseline only, preserving selected dropout. Analytical estimates do not run Monte Carlo for every cell.

Simulation seed

42

Integer · 0–4,294,967,295

Simulation replicates

10,000

Integer · 1–100,000

Run simulation

Not yet run · No simulation estimate available. Power is conditional on assumptions, not the probability that a drug works.

Download current scenario JSON · unavailable

Run a current selected-scenario simulation before exporting a complete record.

Route / · Ready only with matching valid grid point and available service adapter. Capture frozen design, scenario, method/hash, seed 42 and 10,000 replicates on Run. Settings edits invalidate results. Backend progress/error live region; retry same captured context. No estimate until service returns.

Better Baseline

LIVE / SERVICE BINDINGS · INVALID

09 / INVALID · Route /
Illustrative domain-error state; no live calculation.

This scenario cannot be evaluated.

INVALID · analytic power null — not 0%.
Treatment event risk must be positive.
Δ power unavailable; simulation blocked.

LOCKED ENROLLMENT

‹N_C› control
‹N_T› treatment

Enrollment frozen before losses.
Retained: floor(N × (1−0.00)) / arm
Stress edits never change enrollment.

Selected stress scenario

Control endpoint risk

10% / fraction 0.10
1–99% · steps of 1 percentage point

Symmetric dropout

0% / fraction 0.00
0–50% · steps of 1 percentage point

Inherited ARR 0.12 (12pp) · target 80%.
Stress risk 0.10 → treatment risk −0.02.
Outside model domain; revise assumptions.

Evaluate Stress Scenario

Load grid response from backend

Original vs selected

Control risk	Plan: 20% Stress: 10%
Dropout	Plan: 0% planned Stress: 0% selected
Frozen enrollment C / T	Plan: ‹N_C› / ‹N_T› Stress: ‹N_C› / ‹N_T›
Conditional power	Plan: ‹original analytic power› Stress: null · INVALID

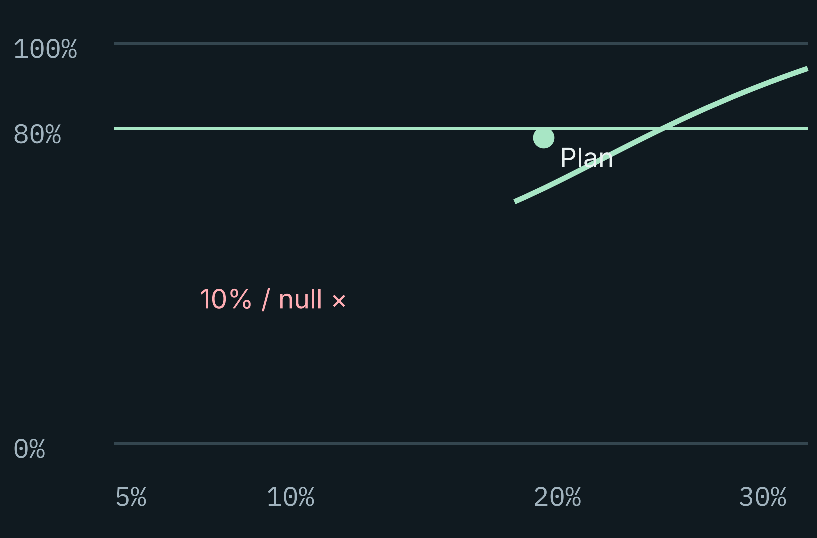
```
Expected retained C / T (floored)
Plan: ‹E_C plan› / ‹E_T plan›
Stress: ‹E_C stress› / ‹E_T stress›
```

Δ power unavailable — never zero

Target 80% · inherited ARR 0.12
Cohen’s-h / normal approximation
Hash ‹matching hash› · version ‹method version›
Warnings ‹model-domain diagnostics›

Conditional power curve

Fixed ‹N_C› / ‹N_T› · dropout exactly 0%.
Schematic gap at invalid baseline; no selected power.



Control baseline risk (%) →
Conditional analytic power (%) ↑
Reference: target 80% · gaps at invalid values

Numerical alternative / select baseline

10% · null × · selection blocked

20% · ‹power› · revise baseline

Invalid values · null × · disabled

Original reference uses original risk and planned dropout. Current point 10% / 0% is invalid: no curve mark, no selection. Gaps stay open; never interpolate over null.

Sensitivity grid

Risk columns 5–30% / dropout rows 0–30%.
Null × at invalid cells; no target boundary across nulls.

Loss↓	5%	10%	20%	30%
0%	null ×	null ×	‹p›	‹p›
null ×	null ×	null ×	‹p›	‹p›
20%	null ×	null ×	‹p›	‹p›
30%	null ×	null ×	‹p›	‹p›

× invalid / null · no selected power
‹p› binds a valid returned power. Target envelope ends at invalid cells.

Invalid cells disabled for keyboard/touch selection and simulation. Valid cells may correct both controls. Numeric grid and CSV preserve nulls, flags, precision, hashes and warnings.

Download analytic grid CSV

Run simulation · blocked: invalid scenario

```
Route / · Announce invalid status and model-
domain reason. No zero substitute, no
interpolation across nulls. Simulation and
complete JSON export blocked. Evaluate Stress
Scenario retries edited inputs; planning
changes require a new design lock.
```

10 / SELECTED-SCENARIO SIMULATION

One scenario. Reproducible execution.

EXECUTION SETTINGS

Seed (integer)

42

0–4,294,967,295 · default 42

Replicates (integer)

10,000

1–100,000 · default 10,000

Requires a valid current LIVE design, selected scenario and available live adapter. Enrollment remains frozen.

RUNNING · Monte Carlo simulation in progress...

Seed 42 · Requested replicates 10,000
Captured control risk 10% · symmetric dropout 0%
Locked control / treatment ‹N_C› / ‹N_T›
Request inputs hash ‹captured hash›

Indeterminate progress: do not invent a completion percentage.

Running simulation...

Download current scenario JSON · unavailable

No current simulation result yet. Preserve entered settings and frozen enrollment. Disable duplicate submission; announce running and completion using a polite live region.

Alternate state · FAILED

The simulation service could not complete this request. Your inputs and enrollment lock are unchanged. Retry when the adapter is available.

Retry simulation

10 / FUNCTION MAP · Polite live region announces running/completion; duplicate submit disabled. Edits invalidate capture. Discard late responses; prior output is stale only. Retry preserves inputs and frozen enrollment.

11 / SELECTED-SCENARIO SIMULATION

Inspect the result, then preserve the record

ILLUSTRATIVE SERVICE-BINDING SPECIFICATION · Current-result UI, not an actual run. ‹Fields› bind returned outputs; no verified numbers or identifiers are fabricated.

EXECUTION SETTINGS

Seed (integer)

42

0–4,294,967,295 · default 42

Replicates (integer)

10,000

1–100,000 · default 10,000

Requires a valid current LIVE design, selected scenario and available live adapter. Enrollment remains frozen.

CURRENT · Design, selection, curve, grid and simulation match

Risk 10% · dropout 0% · RR 0.75 · two-sided α 5% · target 80%
Frozen enrollment ‹N_C› / ‹N_T› · seed 42 · requested replicates 10,000

Selected-scenario Monte Carlo

Two-sided significance fraction

‹two-sided fraction›

MCSE ‹two-sided MCSE› percentage points

Favorable-direction significance fraction

‹favorable fraction›

MCSE ‹favorable MCSE› percentage points

Invalid-test fraction

‹invalid fraction›

‹Undefined tests› / ALL actual replicates. Undefined tests are nonrejections, never omitted.

Actual replicates ‹returned count› · seed ‹returned seed›
Both significance fractions use ALL actual replicates.
Favorable = significant in the lower-adverse-risk treatment direction.

Enrollment is not retention

Frozen enrolled before losses

Control ‹N_C› / Treatment ‹N_T›

Floored expected evaluable · analytic

Control ‹expected_C› / Treatment ‹expected_T›

Mean sampled actual retained · simulation

Control ‹mean retained_C› / Treatment ‹mean retained_T›

Mean simulated events

Control ‹mean events_C› / Treatment ‹mean events_T›

Inflation accounts for expected loss; it does not guarantee retained counts. Monte Carlo samples actual independent retention.

Methods & execution provenance

Method ‹returned method; expected pooled two-proportion z test›
Model version ‹returned version› · RNG ‹returned generator›
Software versions ‹returned environment› · Timestamp ‹returned UTC›
Inputs hash ‹returned hash› · Result identifier ‹returned result ID›

Returned field provenance: source, description and source URL where available for each output. Verification/retrieval metadata preserved; no clinical source invented.

Simulation warnings ‹returned warnings, or explicit none returned›
Invalid tests are nonrejections; analytic and simulation methods are different approximations.

A complete, matching scenario record

Available only after every current input, design, grid/curve and selected simulation matches the displayed seed and replicate count.

Download current scenario JSON

Record contains assumptions, fixed enrollment, stress, analytic design/grid/curve, simulation and separate MCSEs, warnings/provenance, seed/replicates, methods/models/software/RNG, timestamps and hashes.

MCSE measures simulation sampling error, not uncertainty in the assumed rates or biological effect. More replicates improve estimate precision, not a real trial’s success probability.

11 / FUNCTION MAP · CURRENT is a UI state specification. Enable JSON only on complete, matching design/grid/curve/simulation/settings. Download the complete artifact, not the visible summary.

12 / SELECTED-SCENARIO SIMULATION

Changed settings. Previous result is not current.

STALE · Replicate count changed to 20,000

Previous simulation: seed 42 / 10,000 replicates / ‹old inputs hash›. Keep prior estimates only under this stale label; do not use them in current export.

EXECUTION SETTINGS

Seed (integer)

42

0–4,294,967,295 · default 42

Replicates (integer)

20,000

1–100,000 · default 10,000

Requires a valid current LIVE design, selected scenario and available live adapter. Enrollment remains frozen.

Simulation seed: 42
Simulation replicates: 20,000
Current selected scenario: ‹risk› / ‹dropout› · lock ‹N_C› / ‹N_T›

Run simulation

Download current scenario JSON · unavailable

Unavailable: displayed replicate count does not match captured simulation. Run a matching simulation; ensure design and analytic data are current.

INVALID INPUT · Seed −1 / replicates 100,001

Seed must be an integer between 0 and 4,294,967,295. Replicates must be an integer between 1 and 100,000. Correct settings before running.

Run simulation · blocked by invalid settings

Better Baseline

LIVE · Calculation service

The service did not finish. Your work is safe.

Plan a hypothetical trial. Lock enrollment. Stress-test assumptions.

SYNTHETIC PRESET / 3 EXAMPLES

Selected · Baseline collapse

20% → 10% risk · 0% → 0% dropout

`ID/version: {baseline_collapse.identity}`
`Preset hash: {baseline_collapse.hash}`

Attrition shock

20% → 20% risk · 5% → 25% dropout

`ID/version: {attrition_shock.identity}`
`Preset hash: {attrition_shock.hash}`

Stable design

20% → 19% risk · 5% → 6% dropout

`ID/version: {stable_design.identity}`
`Preset hash: {stable_design.hash}`

Lower control event risk, unchanged hypothetical effect. Synthetic assumptions; not clinical evidence.

`All presets: RR 0.75 · target 80% · two-sided α 5% · seed 42 · 10,000 replicates`

Reset to Preset Defaults

Change or reset restores preset assumptions, seed and replicates; invalidates pending work and captured results.

01 / Planning assumptions

Equal allocation · binary adverse outcome · two-sided superiority

Control endpoint risk (%)

20

Enter 20 for 20%. Risk in eligible control participants over the specified follow-up, not prevalence. Valid: >0 and <100.

Hypothetical effect / RR or ARR

RR · ratio | ARR · pp

Switch RR / ARR with arrow keys. Hypothetical effect, not efficacy evidence.

Relative risk (ratio)

0.75

Decimal ratio; strictly >0 and <1.

Target power (%)

80

>0 and <100; must exceed alpha.

Two-sided alpha (%)

5

>0 and <100.

Dropout in each arm (%)

0

One shared value for both arms. Valid: 0 to <100.

Retry calculation

Successful calculation sets a new enrollment lock. Editing never silently replaces an existing lock.

02 / Enrollment lock

Failed · no new lock

The calculation service is unavailable. Your entered assumptions are preserved. Check the local backend connection and retry.

`Unavailable · not zero`

No current power returned.
Existing lock, if any: retained as stale.
Stress evaluation/grid: unavailable until service recovers.

```
Source: synthetic preset
Follow-up: {preset.follow_up_label}
No clinical source asserted.
```

Retry stress evaluation · when valid

`LIVE ≠ clinical evidence`

Fresh numerical results require the calculation service. A local backend can work without a registry or internet connection. Mode is externally configured; no mode toggle.

Load grid response from backend

Grid retry requires a valid current design and an available adapter. Failure leaves stale outputs explicitly labeled, never substituted with zero.

13 / INTERACTION · RECOVERY

Show an understandable service error, preserve inputs and prior lock, and announce failure with role=alert. Offer Retry calculation, Evaluate Stress Scenario, and "Load grid response from backend" retries in their respective failed regions. Never promote fixture or late results to current.

14 / MOBILE / WORKSPACE STATE

Better Baseline

DEMO · Illustrative only

Explore the fixture. No backend execution.

Plan a hypothetical trial. Lock enrollment. Stress-test assumptions.

SYNTHETIC PRESET / 3 EXAMPLES

Selected · Baseline collapse

20% → 10% risk · 0% → 0% dropout

`ID/version: {baseline_collapse.identity}`
`Preset hash: {baseline_collapse.hash}`

Attrition shock

20% → 20% risk · 5% → 25% dropout

`ID/version: {attrition_shock.identity}`
`Preset hash: {attrition_shock.hash}`

Stable design

20% → 19% risk · 5% → 6% dropout

`ID/version: {stable_design.identity}`
`Preset hash: {stable_design.hash}`

Lower control event risk, unchanged hypothetical effect. Synthetic assumptions; not clinical evidence.

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Reset to Preset Defaults

Change or reset restores preset assumptions, seed and replicates; invalidates pending work and captured results.

---

01 / Planning assumptions

Equal allocation · binary adverse outcome · two-sided superiority

Control endpoint risk (%)

20

Enter 20 for 20%. Risk in eligible control participants over the specified follow-up, not prevalence. Valid: >0 and <100.

Hypothetical effect / RR or ARR

RR · ratio | ARR · pp

Switch RR / ARR with arrow keys. Hypothetical effect, not efficacy evidence.

Relative risk (ratio)

0.75

Decimal ratio; strictly >0 and <1.

Target power (%)

80

>0 and <100; must exceed alpha.

Two-sided alpha (%)

5

>0 and <100.

Dropout in each arm (%)

0

One shared value for both arms. Valid: 0 to <100.

Live calculation unavailable · DEMO

Successful calculation sets a new enrollment lock. Editing never silently replaces an existing lock.

---

02 / Enrollment lock

Illustrative fixture · not live

Baseline collapse hand-authored fixture. Not newly calculated, and not a verified simulation result.

```
DEMO fixture · enrollment held fixed
Control enrolled: 903
Treatment enrolled: 903
Stressed power: approximately 47%
Selected risk: 10% · dropout: 0%
```

```
Source: synthetic preset
Follow-up: {preset.follow_up_label}
No clinical source asserted.
```

Run simulation · unavailable in DEMO

`Complete scenario JSON unavailable`

DEMO uses fixtures/client calculations, no backend execution. Monte Carlo cannot run; complete export requires matching live simulation. Analytic grid CSV remains a separate download.

Download analytic grid CSV

Complete scenario JSON · unavailable

---

14 / INTERACTION · DEMO RESTRICTIONS

Synthetic inputs can be used in genuine LIVE calculations; synthetic input is not demo output. Mode is external, with no toggle. Other supplied fixtures only: Attrition shock 951/arm, ≈50% stressed power; Stable design 951/arm, ≈78%. All illustrative; never invent simulation output.

Better Baseline

LIVE · Calculation service

Validate a record, not a restored project.

Plan a hypothetical trial. Lock enrollment. Stress-test assumptions.

SYNTHETIC PRESET / 3 EXAMPLES

Selected · Baseline collapse

20% → 10% risk · 0% → 0% dropout

`ID/version: {baseline_collapse.identity}`
`Preset hash: {baseline_collapse.hash}`

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20% → 19% risk · 5% → 6% dropout

`ID/version: {stable_design.identity}`
`Preset hash: {stable_design.hash}`

Lower control event risk, unchanged hypothetical effect. Synthetic assumptions; not clinical evidence.

`All presets: RR 0.75 · target 80% · two-sided α 5% · seed 42 · 10,000 replicates`

Reset to Preset Defaults

Change or reset restores preset assumptions, seed and replicates; invalidates pending work and captured results.

Scenario record

Export a complete, current record only when all inputs and results match. Validate a previously downloaded file without changing this workspace.

Download current scenario JSON

Export unavailable

No complete matching result: calculate a current design, refresh analytic curve/grid, and run the selected live simulation with the displayed seed and replicate count. Never combine stale results with new inputs.

Record includes assumptions, frozen enrollment, stress, analytic design/curve/grid, simulation outputs, separate MCSEs, warnings, field provenance, seed/replicates, method/model/software versions, RNG, timestamps and hashes.

Reopen and validate scenario JSON

Choose JSON file

Accepts .json · validation only; not patient data

File validation feedback

Valid JSON · detached artifact

Schema: {validated.schema_version}
Export time: {validated.export_time}
Seed: {validated.seed}
Replicates: {validated.replicates}

The artifact matches the schema. Your current assumptions and results are unchanged.

Invalid file · alternate feedback

Schema validation failed: required fields are missing or malformed.
{validation.error_path}: {validation.error_message}
Choose a complete scenario JSON and retry; workspace unchanged.

Choose another JSON file

`These are alternate success/error states, shown together for implementation review. Display only the applicable feedback in the workspace; returned metadata uses service/file bindings.`

15 / INTERACTION · DETACHED VALIDATION

Use a labeled file input accepting JSON. Announce success via status and errors via alert; keep keyboard focus on the validator. Reopening never restores assumptions, replaces the lock, or reruns calculations. Schema validity does not independently prove statistical correctness. Reproduction requires the recorded software environment.

Better Baseline

LIVE / WORKSPACE DISCLOSURE

Trial Planning Workspace / Expanded audit trail

Assumptions, methods & provenance

> STALE / UNHASHED · Manual planning edits do not have a current result hash. Last design hash ‹previous service hash› remains stale.

Field: Control endpoint risk

`Value: 22% · manual edit`

Source / description: Manual-user · hypothetical control event probability

Metadata: No clinical URL · unhashed

Field: Effect model / value

`Value: RR / 0.75`

Source / description: Synthetic preset · hypothetical risk ratio

Metadata: ‹versioned preset identity›

Field: Target / alpha

`Value: 80% / 5%`

Source / description: Synthetic preset · planning thresholds

Metadata: ‹inputs hash› · no clinical source

Field: Planned dropout

`Value: 0% each arm`

Source / description: Synthetic preset · symmetric independent losses

Metadata: ‹available source metadata›

Field: Stress control risk / dropout

`Value: 0.10 / 0.00 fractions`

Source / description: Manual-user · selected stress assumptions

Metadata: No clinical source · selection hash pending

Field: Seed / requested replicates

`Value: 42 / 10,000 integers`

Source / description: Synthetic preset defaults · execution settings

Metadata: Invalidated by edits

Field: Design / analytic / simulation outputs

`Value: ‹returned fields›`

Source / description: Calculation-service field provenance

Metadata: ‹URL if available› · ‹retrieval / verification metadata›

Tracked fields show source, description and available URL. Preset selection/reset restores synthetic provenance, assumptions and simulation defaults and invalidates captured/pending work. Edited fields acquire manual-user provenance. Preserve supplied generator, scenario identity, follow-up label, model/method versions and warnings.

Available source & result metadata

Scenario identity: ‹returned scenario identity›
Follow-up: ‹source follow-up label› · descriptive context only
Generator: ‹supplied generator›
Source URL: ‹available source URL, otherwise not supplied›

Method / model: ‹returned versions›
Result identity / hash: stale · ‹previous identifiers›
Retrieval & verification: ‹supplied metadata›
Current input hash: unavailable after manual edits

What the model means

Baseline risk

A specified binary event in eligible control participants over a stated follow-up period; not general disease prevalence.

Hypothetical treatment risk

Implied by RR or ARR, not evidence of efficacy.

Supported model

Two arms, equal allocation, binary adverse-outcome superiority, two-sided test, independent symmetric dropout.

Analytic method

Cohen’s-h / normal approximation with floored expected retained counts. Conditional power depends on assumptions and method.

Simulation method

Sample actual retained counts; pooled two-proportion z test. This differs from the analytic approximation.

Enrollment and retention

Enrolled counts are before losses. Required evaluable counts, floored expected retained counts and sampled actual retention are different quantities. Inflation for expected losses never guarantees retained counts.

What the model does not claim

MCSE

Simulation sampling error only; not uncertainty in assumed event rates or biological effect.

Replicate precision

More replicates improve simulation precision, not variability or success of a real trial.

Two different rejections

Two-sided and favorable-direction rejection differ; under a symmetric null they are approximately α and $\alpha/2$.

Target-power envelope

Hypothetical assumptions meeting the chosen threshold; not a clinical or regulatory safe zone.

No certification or prediction

Does not predict drug efficacy, FDA approval, trial or commercial success, or cost savings. Does not provide clinical/regulatory design certification.

WARNINGS & REPRODUCIBILITY · ‹Returned design, grid and simulation diagnostics›. Source metadata does not establish clinical truth. Schema-valid JSON does not prove statistical correctness; reproduction requires the recorded software environment.

In-workspace disclosures use labelled disclosure buttons with expanded/collapsed state. Tables have captions and scoped headers; charts have numerical alternatives. Minimum 44px targets, visible keyboard focus, no status conveyed by color alone. Announce evaluation, invalidity, errors and export availability. No mode toggle or extra routes.

STATE 16 / Same route /. Expanded Methods, warnings & field provenance disclosure. Collapsing this region returns to the working surface; it does not navigate. Manual fields stay unhashed until recalculation. Returned metadata is displayed only when supplied.

Collapse methods & provenance –